NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Centre for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI



Module 3 · Machine Learning & GenAI

L05
Unsupervised **Learning**

there’s no target column like churned, no “right answers”

Sarah's week of finding structure without labels

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Where We Left Off

L04 → L05 — from prediction to discovery

Marcus, end of L04:

"Now — what about the customers who DON'T churn but also don't log in for a year? Can we find natural CLUSTERS of customer behaviour without labels?"

What's different this week

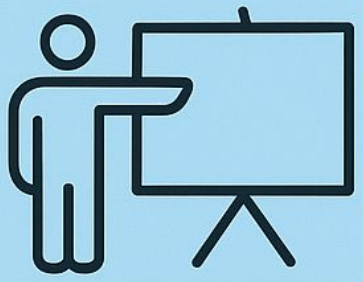
📂 **Same dataset, no label**	10,000 NorthStar customers — `churned` column dropped
🎯 **No 'right answer'**	We're looking for structure, not predicting a target
🛠️ **Three new tools**	PCA (compress) · K-Means (group) · Isolation Forest (flag)
🏁 **Friday deliverable**	Segments + anomaly list for marketing and customer success

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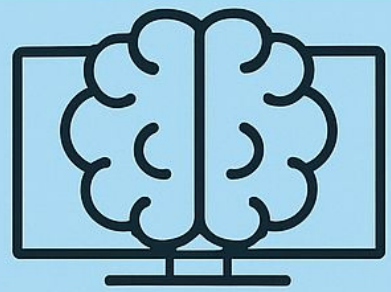
MACHINE LEARNING



SUPERVISED LEARNING



UNSUPERVISED LEARNING



- Regression
- Classification
- Anomaly Detection
- Dimension Reduction
- Clustering

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Real-World Applications of Unsupervised Learning

Clustering (Discovering Hidden Groups)

- Customer Segmentation: Grouping users for targeted marketing.
- Content & Topic Categorization: Grouping text into themes.
- Biological Data Analysis: Grouping genes/patients for disease subtypes.

Anomaly Detection (Spotting the Unusual)

- Fraud Prevention: Flagging unusual transactions.
- Predictive Maintenance: Analyzing sensor data for equipment risks.
- Cybersecurity: Identifying unusual network traffic.

Dimensionality Reduction (Simplifying Complexity)

- Data Visualization: Compressing data to 2D/3D to spot trends.
- Image & Video Compression: Reducing file size while keeping core structure.
- Feature Extraction: Simplifying data to speed up models and reduce noise.

Association Rules (Finding Relationships)

- Market Basket Analysis: Powering recommendation engines via purchasing connections.

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Three Categories of Unsupervised Work

One algorithm from each — covers most of industry



Dimensionality reduction

Squeeze high-dim data into fewer dimensions without losing the signal.

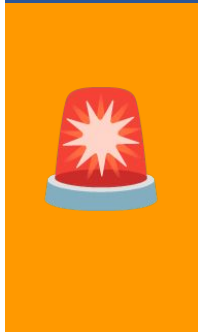
▸ **PCA**



Clustering / segmentation

Group similar items together.

▸ **K-Means**



Anomaly detection

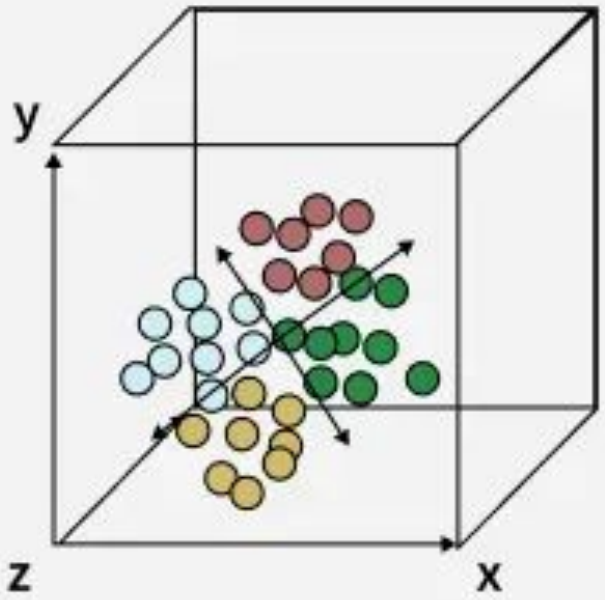
Flag items that look unlike everything else.

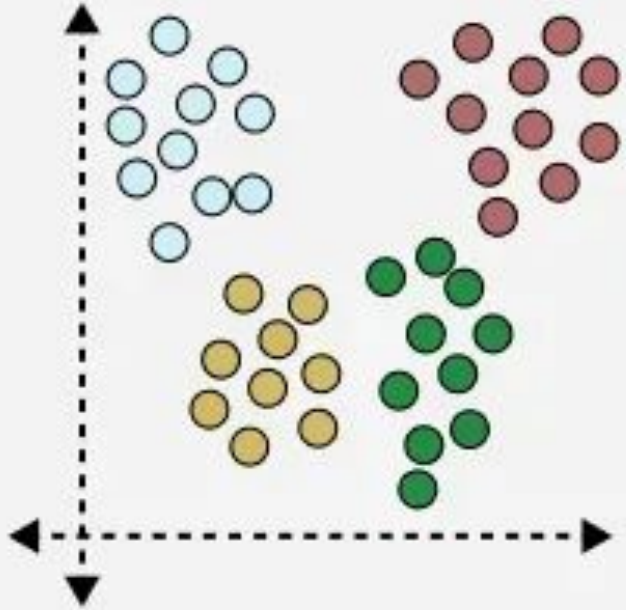
▸ **Isolation Forest**

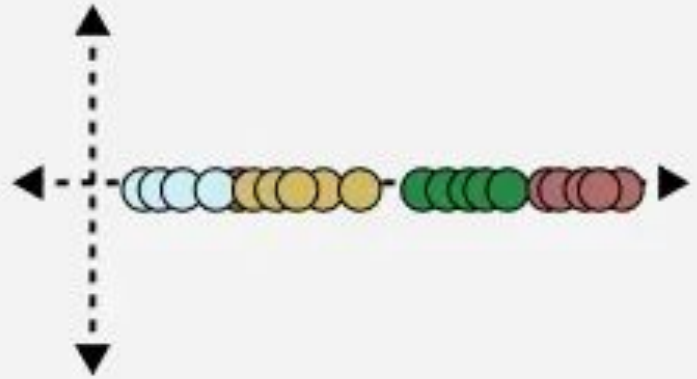
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What is Dimensionality Reduction

Dimensionality Reduction is the process of reducing the number of input variables (features) in a dataset while preserving as much important information as possible.







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Why we need Dimensionality Reduction?

- **Data Visualization:** Compresses high-dimensional data (e.g., 50 features) into 2D or 3D to easily spot patterns, clusters, and outliers in charts.
- **Faster Computation:** Reduces variables to decrease processing complexity, allowing machine learning models to train and run faster.
- **Removing Noise:** Eliminates irrelevant variables or duplicate info while retaining strong signals, preventing overfitting and improving generalization.
- **Saving Storage Space:** Condenses data to reduce memory and server requirements, making storage cheaper and more manageable.

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PCA — Principal Component Analysis

Find new axes that capture the most variance

A loading is the weight each original feature gets inside a component.

The idea

"Find new axes — PC1, PC2, … a weighted linear combination of all your original features. PC1 is the direction in which the data spreads out the most; PC2 is the next most-spread direction that is uncorrelated with PC1, and so on."

When to reach for it

- **Visualisation** — Project 10D customer data into 2D for a chart
- **Preprocessing** — Feed a 5-component representation into a downstream model
- **Decorrelation** — PCs are orthogonal — useful before some algorithms

⚠️ **ALWAYS standardise features before PCA — variance is scale-dependent.**

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PCA Loadings — Naming the Axes

What each principal component actually represents

The size (ignoring the sign) shows how much a feature matters to that component: about 0.4–0.6 is strong, near 0 is negligible. The sign shows the direction: + means a higher feature value pushes the component score up, and − means it pushes it down.

Feature	PC1	PC2
support_tickets_quarter	+0.555	+0.009
age	+0.452	−0.405
avg_review_polarity	−0.432	+0.113
tenure_months	+0.352	−0.239
last_login_days_ago	+0.307	+0.468
returns_per_purchase	−0.233	−0.437
num_purchases_quarter	+0.126	+0.591

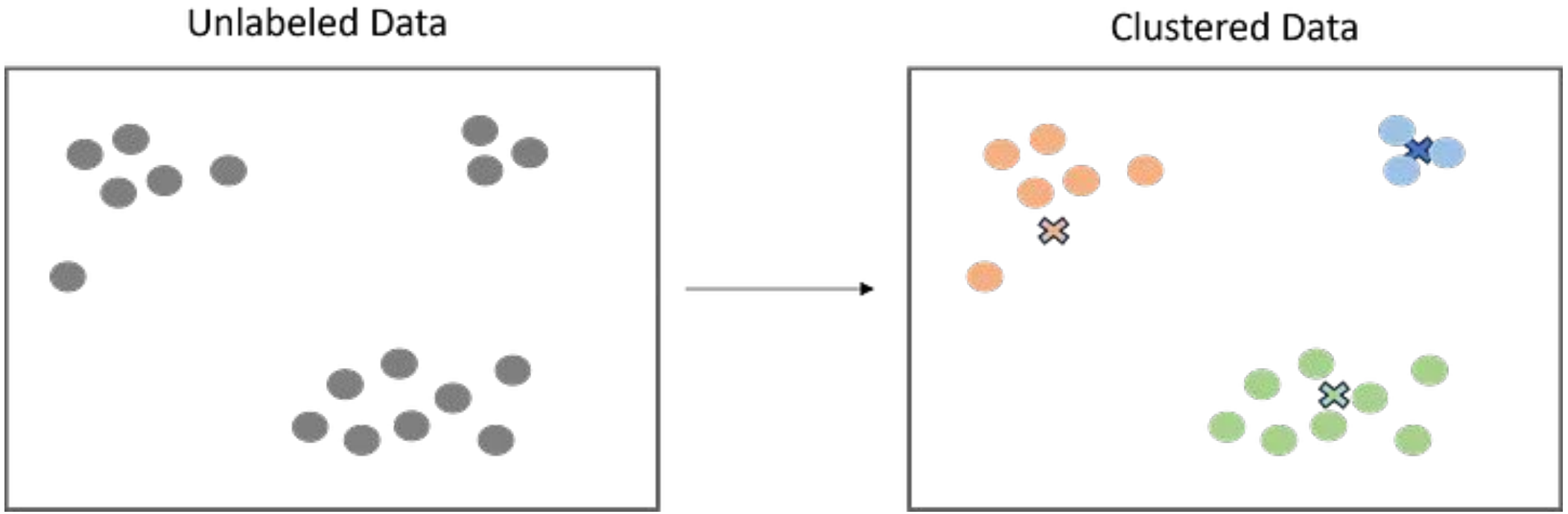
Sarah's interpretation

PC1 → 'frustration': high tickets, age, low reviews. PC2 → 'activity': high purchases, low returns.

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K-Means

K-Means asks “which centre are you nearest to?”
DBSCAN asks “are you connected to a crowd?”



K-Means autonomously organizes unlabelled data by grouping similar data points into distinct clusters without requiring predefined categories.

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K-Means — The Standard Clustering Algorithm

Partition data into K groups by minimising within-cluster variance

The algorithm (Lloyd's algorithm)

1. Pick K starting centroids (randomly or with k-means++)
2. Assign each point to its nearest centroid
3. Move each centroid to the mean of its assigned points
4. Repeat from step 2 until centroids stop moving

When K-Means works

- ✓ Compact, roughly spherical clusters
- ✓ Moderate dimensionality (run PCA first if you have > 50 features)
- ✓ Reasonably balanced cluster sizes
- ✗ Non-globular shapes (moons, rings) — use DBSCAN instead

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Choosing K — Three Approaches

Mathematics + judgement

K-Means needs you to choose k, the number of clusters, before it runs.

1

Elbow method

A kink is just another word for the elbow: the point on the inertia chart where the line bends sharply, going from a steep drop to a flat line. We pick the k where the line starts to flatten because that's the point where adding more clusters stops finding real groups and starts just chopping up groups that already exist

Plot inertia vs K. Look for the 'bend' where adding more K stops helping much.

- **Run K=2..10. Plot. Find the kink.**

Inertia is a number that measures how tightly packed the clusters are. Specifically, it's the total of the squared distances from each point to the centre of its own cluster. Low inertia means points sit close to their centres, so the clusters are tight and compact. High inertia means points are spread far from their centres, so the clusters are loose.

2

Silhouette score

Per-point measure of cluster fit, averaged. Higher = better.

- **Peak score → that K is mathematically the best.**

Where inertia only checks tightness, the silhouette checks two things at once: are clusters tight, and are they well separated from each other?

3

Business judgement

Are 4 segments more actionable than 7? Marketing teams have limits.

- **Don't ignore operational reality.**

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Cluster Profiling — From Numbers to Names

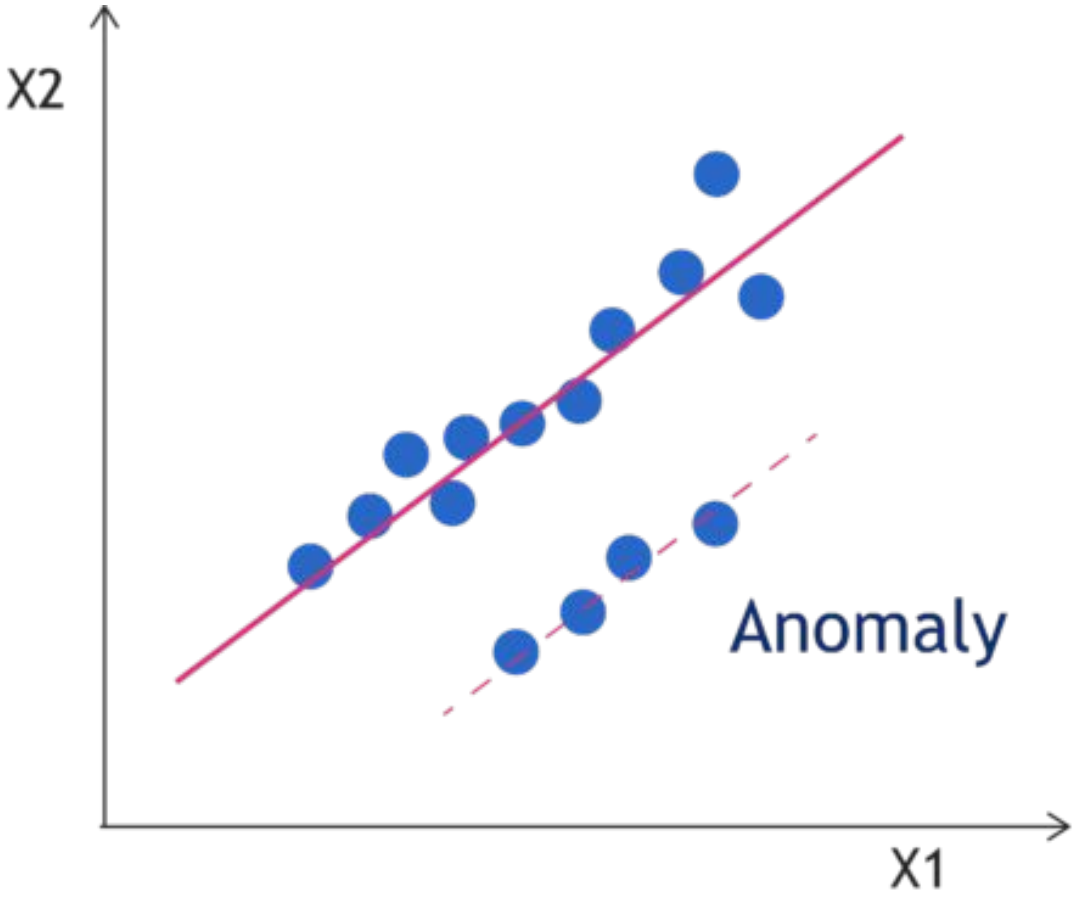
What makes each cluster distinctive?

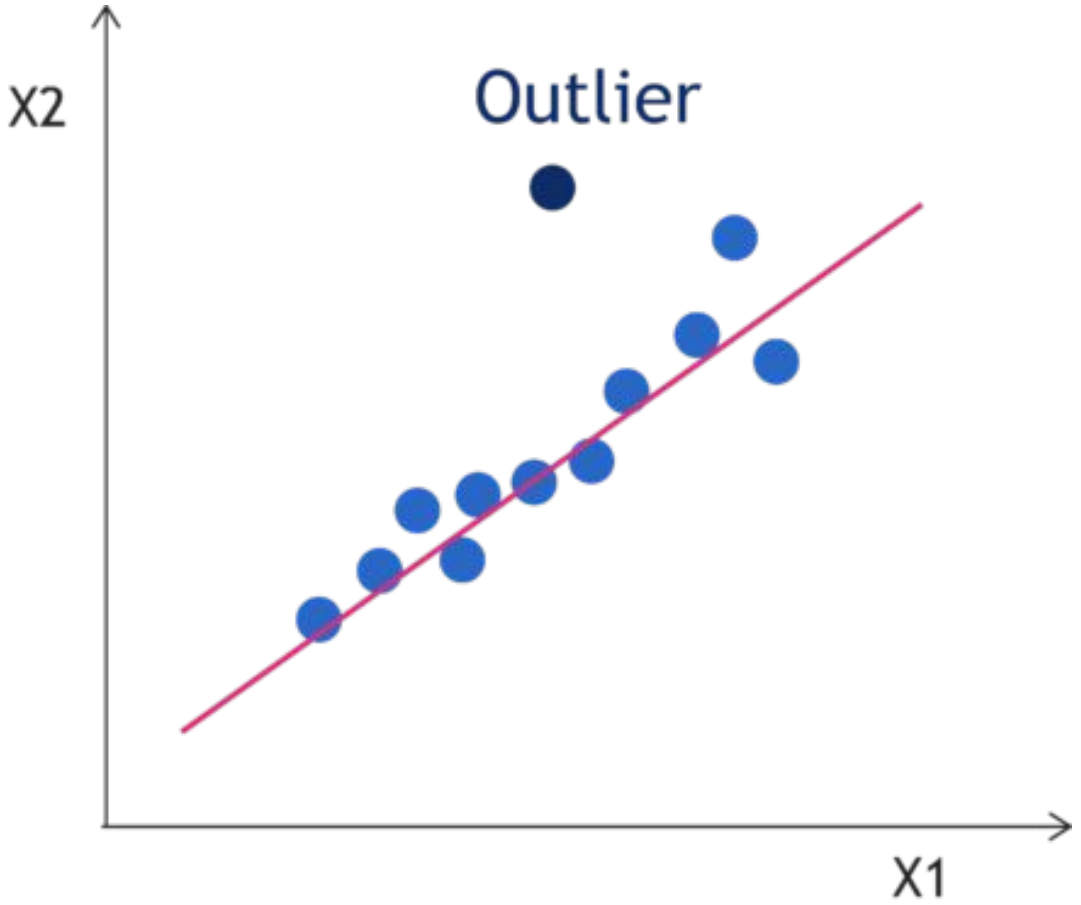
Cluster	tenure	spend	returns	last login	polarity	Sarah's name
Cluster 0	55 mo	£71	0.09	21 d	+0.16	**Loyal long-term**
Cluster 1	36 mo	£71	0.13	92 d	+0.12	**Dormant**
Cluster 2	17 mo	£69	0.10	21 d	+0.13	**New & active**
Cluster 3	36 mo	£69	0.31	22 d	+0.18	**High returns**

The Sarah → Marcus translation

"Cluster 0 isn't a number. It's our LOYAL LONG-TERM segment — 35% of our base, low return rate, the people we keep happy.
Cluster 1 are the DORMANT — they haven't logged in for three months on average. That's the re-engagement target."

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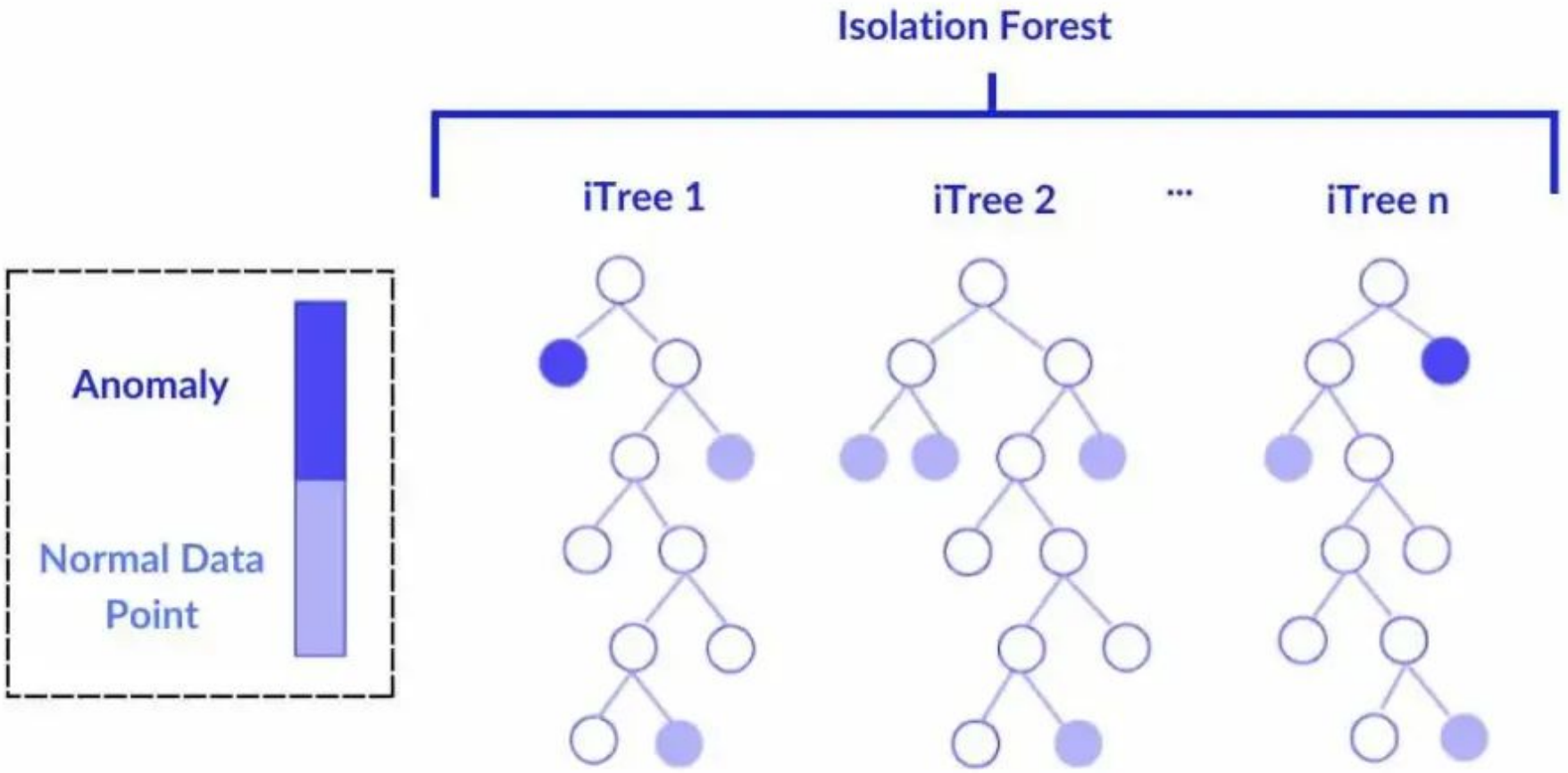




outliers are often viewed as mere data errors or noise to be filtered out

anomalies highlight meaningful, unusual patterns that signal significant events requiring attention.

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- **Anomalies (Dark blue nodes):** Distinct from the data, they separate easily with short paths near the top of the trees.
- **Normal Data Points (Light blue nodes):** Densely clustered, they require many splits, ending up deep at the bottom with longer paths.

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Anomaly Detection — Find Unusual Individuals

Different question from clustering — same data

Clustering asks

"What GROUPS does my data fall into?"

→ Output: K labels, one per point

→ Tool: K-Means, hierarchical, DBSCAN

Anomaly detection asks

"Which INDIVIDUAL points are unusual?"

→ Output: score per point

→ Tool: Isolation Forest, One-Class SVM, autoencoders

Common use cases

🚨 Fraud detection (banking, e-commerce, insurance)
🔧 Manufacturing defects
📊 Data quality (corrupt rows, outlier values)
🌐 Security / intrusion detection

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Isolation Forest — How It Works

Random trees isolate normal points; anomalies get isolated fast

The intuition

"Build random splits in the data. NORMAL points sit in the dense middle — many splits before they're isolated.
ANOMALIES sit at the edges — they're isolated in just a few splits."

```
from sklearn.ensemble import IsolationForest

iso = IsolationForest(
    n_estimators=100,
    contamination=0.05,        # expected fraction of anomalies
    random_state=42,
    n_jobs=-1,
)
labels = iso.fit_predict(X_scaled)    # -1 = anomaly, +1 = normal
scores = iso.score_samples(X_scaled)  # raw scores (lower = more anomalous)
```

contamination sets the threshold only. The continuous score is what matters operationally.

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Three Things to Remember

Beyond the algorithms — the practical wisdom

1

Unsupervised methods FIND structure — humans NAME it

K-Means gives you cluster 0, 1, 2, 3. Sarah translates to 'Loyal', 'Dormant', 'New', 'High return'. The names are the deliverable.

2

ALWAYS scale before PCA / K-Means

Both algorithms are variance/distance-based. Unscaled features will be dominated by the largest-range column. StandardScaler is the right default.

Rescale your features so they’re on a comparable scale, otherwise the column with the biggest numbers takes over

3

Anomaly detection surfaces, humans investigate

The model can't tell you WHY a customer is anomalous. Look at the feature combinations and write the why yourself.

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Assignment Preview

E-commerce segmentation + credit-card anomalies

Part A — StyleHub Segments

`notebooks/assignment.ipynb · Part A`

- 8,000 e-commerce customers, 7 features
- Exercise A1: preprocessor + 2D PCA
- Exercise A2: choose K with elbow + silhouette
- Exercise A3: profile + name segments
- Deliverable: 4-6 named segments

Part B — Card Fraud Anomalies

- 5,000 credit-card transactions
- Exercise B1: fit Isolation Forest
- Exercise B2: surface the top 50 most unusual
- Deliverable: top-50 watch list + recommendation

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Bridge to Lesson 6

Marcus, after the Friday presentation:

"Great. Segments shipped. Watch list assigned.

Now — our sales are SEASONAL. We need to know quarterly revenue 90 days ahead so ops can plan inventory.

Can you forecast?"

Lesson 6 — Time Series Forecasting

Classical decomposition · ETS · ARIMA · ML-based forecasting (LightGBM with lag features)

Different toolkit. Same Sarah-week structure.

Before L06 — finish the L05 assignment in notebooks/assignment.ipynb.

See You in L06

Time Series Forecasting — ETS · ARIMA · ML-based forecasting

Before then: finish the L05 assignment (StyleHub + credit-card)

Or dig into Hierarchical / DBSCAN / UMAP in optional_extensions.ipynb